VGI

Attach a worker. Query its capabilities.

Install & load

```
INSTALL vgi FROM community;
LOAD vgi;
```

01 At a glance

Vector Gateway Interface connects DuckDB to workers written in other languages. A worker advertises catalogs, tables and functions; attach a catalog and query it alongside local data over a subprocess or HTTP connection.

Worker code → Arrow IPC → DuckDB SQL

02 Worker access & inspection

`vgi_catalogs(location)`

Discover catalog names, options and versions before attaching.

`ATTACH catalog (TYPE vgi, LOCATION …)`

Select the advertised catalog and the worker that serves it.

`vgi_table_function(worker, fn, args)`

Call a worker table function directly; args is a LIST.

`vgi_oauth_identity()`

Inspect authenticated identity for attached catalogs.

`vgi_worker_pool()`

List reusable subprocess workers and their process identifiers.

`vgi_worker_pool_stats()`

Inspect worker reuse through per-path hits and misses.

`CALL vgi_clear_cache()`

Clear cached catalog metadata for attached VGI catalogs.

03 Keep in mind

Catalog name and location differ

The ATTACH identifier is a catalog advertised by the worker. LOCATION identifies its URL or executable. Replace example catalog, schema and table names with discovery results. Pushdown and writable tables depend on the worker's capabilities.

04 Discover a hosted worker

REQUIRES SETUP

```
SELECT catalog, source_url
FROM vgi_catalogs(
  'https://worker.example.com/vgi/'
);
```

Requires a compatible VGI build; replace the URL with your worker endpoint.

05 Attach and query

REQUIRES SETUP

```
ATTACH 'analytics' AS app (
  TYPE vgi,
  LOCATION 'https://worker.example.com/vgi/'
);
SHOW TABLES FROM app.main;
SELECT * FROM app.main.events
LIMIT 10;
DETACH app;
```

Assumes the worker advertises analytics with a main.events table.

06 Launch a local worker

MANUAL SETUP

```
ATTACH 'example' AS local_worker (
  TYPE vgi, LOCATION './my_worker',
  pool true
);
SELECT * FROM vgi_worker_pool_stats();
DETACH local_worker;
```

Requires a trusted executable VGI worker advertising the example catalog.

07 In practice

Choose where code runs

Local workers execute with your process privileges; use workers you trust. HTTP workers run in their own environment and may require OAuth. Browser builds use HTTP transport. Community extension availability varies by DuckDB version and platform.

Full docs & examples ↗ Source ↗